

AP EAMCET-2019-Engineering-21 Apr-Shift-1
Questions physic

81. Two intervals of time are measured as $\Delta t_1 = (2.00 \pm 0.02) \text{ s}$ and $\Delta t_2 = (4.00 \pm 0.02) \text{ s}$. The value of $\sqrt{(\Delta t_1)(\Delta t_2)}$ with correct significant figures and error is
- (1) $(2.828 \pm 0.01) \text{ s}$
 - (2) $(2.83 \pm 0.01) \text{ s}$
 - (3) $(2.828 \pm 0.0075) \text{ s}$
 - (4) $(2.83 \pm 0.0075) \text{ s}$
82. The speed of the particle changes from $\sqrt{5} \text{ ms}^{-1}$ to $2\sqrt{5} \text{ ms}^{-1}$ in a time t . If the magnitude of change in the velocity is 5 ms^{-1} , the angle between the initial and final velocities of the particle is
- (1) 30°
 - (2) 45°
 - (3) 60°
 - (4) 90°
83. The maximum height and range of the projectile are 3 m and 4 m respectively then the velocity of the projectile is

(Acceleration due to gravity = 10 ms^{-2})

(1) $20\sqrt{\frac{6}{5}} \text{ ms}^{-1}$

(2) $10\sqrt{\frac{3}{2}} \text{ ms}^{-1}$

(3) $10\sqrt{\frac{2}{3}} \text{ ms}^{-1}$

(4) $20\sqrt{\frac{5}{6}} \text{ ms}^{-1}$

84. A body is projected at an angle other than 90° with the horizontal with same velocity. If the time of ascent of the body is 1 s then the maximum height it can reach is

(Acceleration due to gravity = 10 ms^{-2})

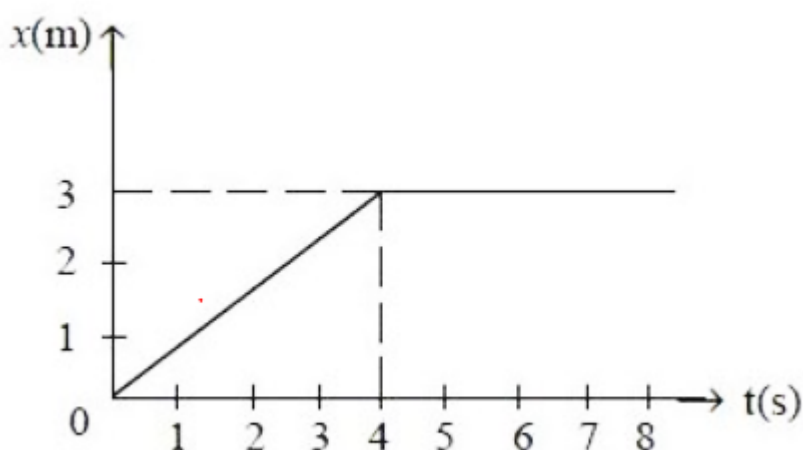
(1) 5 m

(2) 10 m

(3) 2.5 m

(4) 75 m

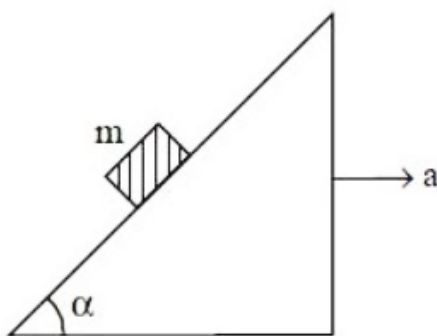
85. The position – time ($x-t$) graph of a moving body of mass 2 kg is shown in the figure. The impulse on the body at $t = 4\text{s}$ is



- (1) $1.5 \text{ kg} \cdot \text{ms}^{-1}$
- (2) $-1.5 \text{ kg} \cdot \text{ms}^{-1}$
- (3) $1 \text{ kg} \cdot \text{ms}^{-1}$
- (4) $2 \text{ kg} \cdot \text{ms}^{-1}$

86. A block of mass ' m ' is lying on a rough inclined plane having an inclination $\alpha = \tan^{-1}\left(\frac{1}{5}\right)$. The inclined plane is moving horizontally with constant acceleration of $a = 2 \text{ ms}^{-2}$ as shown in the figure. The minimum value of the coefficient of friction, so that the block remains stationary with respect to the inclined plane is

(Acceleration due to gravity $= 10 \text{ ms}^{-2}$)



(1) $\frac{2}{9}$

(2) $\frac{5}{12}$

(3) $\frac{1}{5}$

(4) $\frac{2}{5}$

87. Potential energy of a body of mass 1 kg free to move along the X-axis is given by $U(x) = \left(\frac{x^2}{2} - x \right) \text{ J}$. If the total mechanical energy of the body is 2 J, then the maximum speed of the body is (Assume only conservative force acts in the body)

(1) $\sqrt{5} \text{ ms}^{-1}$

(2) 5 ms^{-1}

(3) 3.5 ms^{-1}

(4) $\sqrt{8} \text{ ms}^{-1}$

88. A cylindrical well of radius 2.5 m has water up to a height of 14 m from bottom. If the water level is at a depth of 6 m from the top of the well, then time taken in minutes to empty the well using a motor of 10 HP is approximately

(Acceleration due to gravity = 10 ms^{-2})

- (1) 30
- (2) 80
- (3) 98
- (4) 90

89. A fly wheel of mass 1 kg and radius vector $(2\hat{i} + \hat{j} + 2\hat{k})$ m is at rest. When a force $(3\hat{i} + 2\hat{j} - 4\hat{k})$ N acts on it tangentially, it can rotate freely. Then. Its angular velocity after 4.5 s is

- (1) $\frac{2}{9}\sqrt{261} \text{ rad s}^{-1}$
- (2) $\frac{3}{2}\sqrt{261} \text{ rad s}^{-1}$
- (3) $\sqrt{261} \text{ rad s}^{-1}$

(4) $\frac{5}{9}\sqrt{261} \text{ rad s}^{-1}$

90. Three identical spheres each of diameter $2\sqrt{3}$ are kept on a horizontal surface such that each sphere touches the other two spheres. If one of the sphere is removed, then the shift in the position of the centre of mass of the system is

(1) 12 m

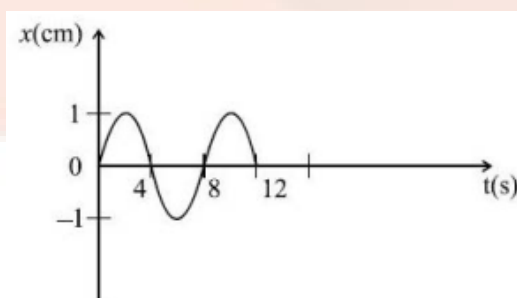
(2) 1 m

(3) 2 m

(4) $\frac{3}{2}$ m

91. For a particle executing simple harmonic motion, the displacement – time ($x-t$) graph is as shown in the figure. The

acceleration of the particle $t = \frac{4}{3}$ s is



$$(1) -\frac{\sqrt{3}}{32}\pi^2 \text{ cm s}^{-2}$$

$$(2) -\frac{32}{\sqrt{3}}\pi^2 \text{ cm s}^{-2}$$

$$(3) +\frac{\sqrt{3}}{32}\pi \text{ cm s}^{-2}$$

$$(4) +\frac{32}{\sqrt{3}}\pi \text{ cm s}^{-2}$$

92. Two masses of 90 kg and 160 kg are separated by a distance of 5 m. The magnitude of intensity of the gravitational field at a point which is at a distance 3 m from the 90 kg mass and 4m from the 160 kg mass is

(Universal gravitational constant = $6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$).

$$(1) 94.3 \times 10^{-10} \text{ N kg}^{-1}$$

$$(2) 9.43 \times 10^{-10} \text{ N kg}^{-1}$$

$$(3) 9.43 \times 10^{-12} \text{ N kg}^{-1}$$

$$(4) 94.3 \times 10^{-12} \text{ N kg}^{-1}$$

93. The following four wires are made of the same material. If same tension is applied to each, the wire having largest extension is

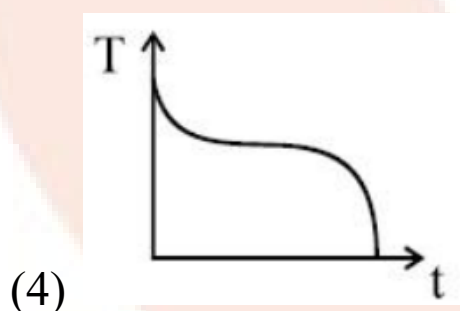
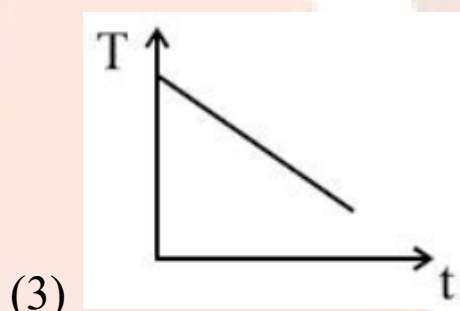
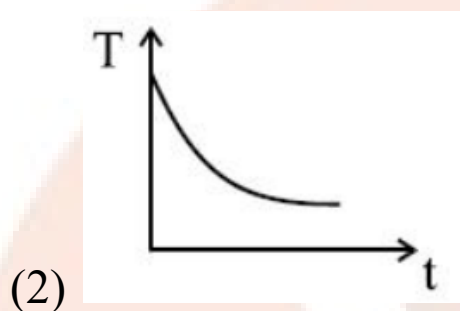
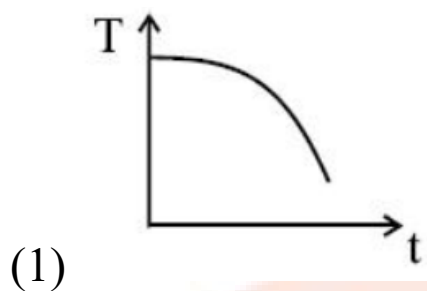
- (1) length 0.5 m, diameter 0.5 mm.
- (2) length 1 m, diameter 1 mm.
- (3) length 2 m, diameter 2 mm.
- (4) length 3 m, diameter 3 mm.

94. A liquid drop of density ρ is floating half immersed in a liquid of surface tension S and density $\frac{\rho}{2}$. If the surface tension S of the liquid is numerically equal to 10 times of acceleration due to gravity, then the diameter of the drop is :

- (1) $\sqrt{\frac{20}{\rho}}$
- (2) $\sqrt{\frac{80}{\rho}}$
- (3) $\sqrt{\frac{60}{\rho}}$
- (4) $\sqrt{\frac{40}{\rho}}$

95. The block of metal is heated to a temperature much higher than the room temperature and placed in an evacuated cavity. The curve which correctly represents the rate of cooling

(T is temperature of the block and t is the time)



96. A solid copper sphere of density ρ , specific heat capacity C and radius r is initially at 200 K. It is suspended inside a chamber whose walls are at 0 K. The time required (in μs) for the temperature of the sphere to drop to 100 K is

(σ is Stefan's constant and all the quantities are in SI units)

$$(1) \ 48 \frac{r\rho C}{\sigma}$$

$$(2) \ \frac{1}{48} \frac{r\rho C}{\sigma}$$

$$(3) \ \frac{27}{7} \frac{r\rho C}{\sigma}$$

$$(4) \ \frac{7}{27} \frac{r\rho C}{\sigma}$$

97. Match the temperatures of the source and sink (T_1 and T_2 respectively) of a Carnot heat engine given in List-I with the corresponding efficiencies in List-II.

List-I

List-II

(A) $T_1 = 500 \text{ K}, T_2 = 300 \text{ K}$

(I) 0.2

(B) $T_1 = 500 \text{ K}, T_2 = 350 \text{ K}$

(I) 0.3

(C) $T_1 = 800 \text{ K}, T_2 = 400 \text{ K}$

(III) 0.4

(D) $T_1 = 450 \text{ K}, T_2 = 360 \text{ K}$

(IV) 0.5

The correct answer is

A

B

C

D

(1) (III)

(IV)

(II)

(I)

(2) (IV)

(III)

(II)

(I)

- | | | | |
|-----------|------|------|------|
| (3) (III) | (I) | (IV) | (II) |
| (4) (III) | (II) | (IV) | (I) |

98. A hammer of mass 200 kg strikes a steel block of mass 200 g with a velocity 8 ms^{-1} . If 23% of the energy is utilized to heat the steel block, the rise in temperature of the block is (specified heat capacity of the steel = $460 \text{ J kg}^{-1} \text{ K}^{-1}$)

- (1) 8°C
- (2) 16°C
- (3) 12°C
- (4) 24°C

99. At the temperature of 314 K and a pressure of 100 kPa, the speed of the sound in a gas is 1380 ms^{-1} . The radius of each gas molecule is $0.5 \text{ \AA}$. The frequency of sound wave at which the wavelength of sound wave in the gas becomes equal to the mean free path of the gas molecules is

(Boltzmann constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$)

- (1) 1000 MHz
- (2) $1000\sqrt{2} \text{ MHz}$

(3) $\frac{1000}{\sqrt{2}}$ MHz

(4) 500 MHz

100. At a temperature of 27°C , two identical organ pipe produces a note of frequency 140 Hz. If the temperature of one pipe is raised to 57.75°C , then the number of beats produced per second is

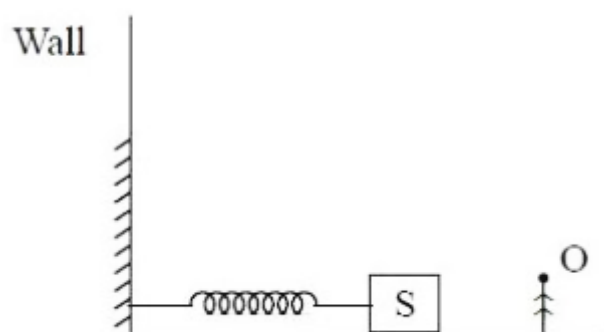
(1) 7

(2) 5

(3) 3

(4) 9

101. A source of sound S in the form of a block kept on smooth horizontal surface is connected to a spring as shown in the figure. If the spring oscillates with an amplitude of 50 cm along the horizontal between the wall and the observer O, the maximum frequency heard by the observer is 12.5 % more than the minimum frequency heard by him. If the mass of the source of sound is 100 g, the force constant of the spring is (Speed of sound in air is 340 ms^{-1})



- (1) 40 N m^{-1}
- (2) 80 N m^{-1}
- (3) 160 N m^{-1}
- (4) 320 N m^{-1}

102. A girl of height 150 cm with her eye level at 140 cm stands in front of plane mirror of height 75 cm fixed to a wall. The lower edge of the mirror is at a height of 85 cm above her feet level. The height of her image the girl can see the mirror is

- (1) 130 cm
- (2) 140 cm
- (3) 120 cm
- (4) 150 cm

103. Unpolarised light from the air incidents on the surface of as transparent medium of refractive index 1.414 such that the

reflected light is completely polarised. Match the angle given in the List-I with the corresponding values given in the List-II.

List-I

List-II

(A) Angle of reflection

(I) $2\sin^{-1}\left(\sqrt{\frac{2}{3}}\right)$

(B) Angle of refraction

(II) $2\sin^{-1}\left(\sqrt{\frac{2}{3}}\right) - \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) Angle between incident and completely polarised rays

(III) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(D) Angle of deviation of the incident ray

(IV) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

The correct answer is

A	B	C	D
(1) (II)	(III)	(I)	(IV)
(2) (II)	(III)	(IV)	(I)
(3) (IV)	(I)	(III)	(II)
(4) (IV)	(III)	(I)	(II)

104. The electric field intensity at a point on the axis of an electric dipole in air is 4 NC^{-1} . Then the electric field intensity at a point on the equatorial line which is at the distance equal to twice the

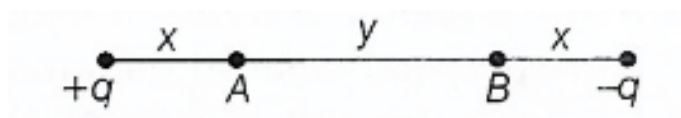
distance on the axial line and if the dipole is in a medium of dielectric constant 4 is

- (1) 1 NC^{-1}
- (2) $\frac{1}{8} \text{ NC}^{-1}$
- (3) 16 NC^{-1}
- (4) $\frac{1}{16} \text{ NC}^{-1}$

105. Two small spheres of each charge ' q ', mass ' m ' and material density ' d ' are suspended from a fixed point with the help of inextensible light thread. When the spheres are in air, the angle between the threads is 90° . When the spheres are suspended in a liquid of density $\frac{2}{3}d$, the angle between the threads is 60° . The value of dielectric constant of the liquid is

- (1) $6\sqrt{3}$
- (2) $2\sqrt{5}$
- (3) $5\sqrt{3}$
- (4) $7\sqrt{2}$

106. The potential difference ($V_A - V_B$) in the arrangement shown in the figure is ($|q| = 1 \mu\text{C}$, $x = 2 \text{ cm}$, $y = 3 \text{ cm}$)



(1) $5.4 \times 10^5 \text{ V}$

(2) $2.7 \times 10^5 \text{ V}$

(3) $5.4 \times 10^2 \text{ V}$

(4) $2.7 \times 10^2 \text{ V}$

107. In a parallel plate capacitor the separation between the plates is $3x$. This separation is filled by two layers of dielectrics, in which one layer has thickness x and dielectric constant $3K$, the other layer is of thickness $2x$ and dielectric constant $5K$. If the plates of the capacitor are connected to a battery, then the ratio of potential difference across the dielectric layers is

(1) $\frac{1}{2}$

(2) $\frac{4}{3}$

(3) $\frac{3}{5}$

(4) $\frac{5}{6}$

108. Assertion (A) : When a wire of aluminium and another wire of silicon are heated from room temperature to 80°C , the conductivity of aluminium increases and that of silicon decreases.

Reason (R) : Aluminium has positive temperature coefficient of resistivity and silicon has negative temperature coefficient of resistivity.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is not correct.

(4) (A) is not correct but (R) is correct.

109. The walls of a closed cubical box of edge 60 cm are made of material of thickness 1 mm and thermal conductivity $4 \times 10^{-4} \text{ cal s}^{-1} \text{ cm}^{-1} \text{ }^{\circ}\text{C}^{-1}$. The interior of the box is maintained 1000°C above the outside temperature by a heater placed inside the box and connected across 400 V DC supply. The resistance of the heater is

- (1) 4.41Ω
- (2) 44.1Ω
- (3) 0.441Ω
- (4) 441Ω

110. A galvanometer of resistance $G \Omega$, is shunted by resistance $S \Omega$. To keep the main current in the circuit unchanged, the resistance to be connected in series with the galvanometer is

- (1) $\frac{G^2}{S + G}$
- (2) $\frac{S}{S + G}$
- (3) $\frac{S^2}{S + G}$
- (4) $\frac{SG}{S + G}$

111. A proton and an α -particle are simultaneously projected in opposite direction into a region of uniform magnetic field of 2 mT perpendicular to the direction of the field. After some time it is found that the velocity of proton has changed in direction by

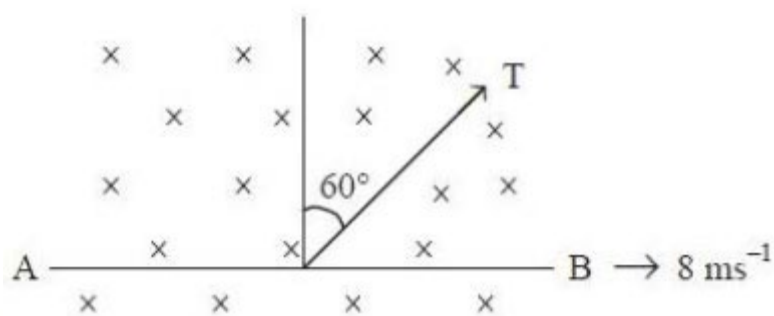
90° . Then at this time, The angle between the velocity vectors of proton and α particle is

- (1) 60°
- (2) 90°
- (3) 45°
- (4) 180°

112. A bar magnet is placed in a uniform magnetic field making an angle θ with the field experiences a torque. If the angle made by the magnet with the field is doubled, the torque experienced by the magnet increases by 41.4 %. The initial angle made by the magnet with the magnetic field is

- (1) 60°
- (2) 30°
- (3) 90°
- (4) 45°

113. A metal rod AB of length 50 cm is moving at a velocity of 8 ms^{-1} in a magnetic field of 2 T. If the field is at 60° with the plane of motion as shown in the figure, then the potentials V_A and V_B are related by



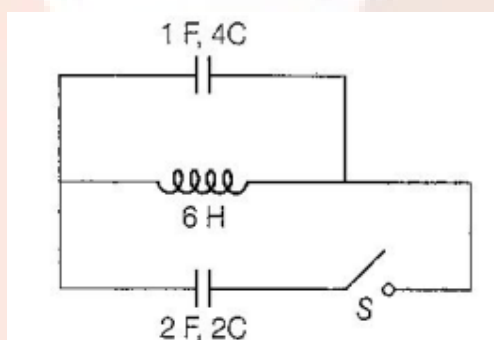
(1) $V_A - V_B = 8 \text{ V}$

(2) $V_A - V_B = 4 \text{ V}$

(3) $V_B - V_A = 8 \text{ V}$

(4) $V_B - V_A = 4 \text{ V}$

114. In the given electrical circuit, if the switch S is closed then the maximum energy stored in the inductors is :



(1) 3 J

(2) 9 J

(3) 12 J

(4) 6 J

115. Which of the following is/are the property/properties of a monochromatic electromagnetic wave propagating in the free space?

- (a) Electric and magnetic field will have phase difference $\frac{\pi}{2}$.
 - (b) The energy of the wave is distributed equally between electric and magnetic fields.
 - (c) The pressure exerted by the wave is the product of its speed and energy density.
 - (d) The speed of the wave is equal to the ratio of the magnetic field to the electric field.
- (1) (a) and (c)
 - (2) only (b)
 - (3) (b) and (c)
 - (4) only (d)

116. The maximum kinetic energy of a photoelectron liberated from the surface of lithium with the work function 2.35 eV by electromagnetic radiation whose electric component varies with the time as:

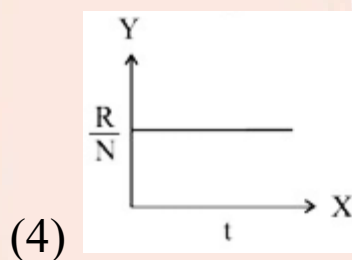
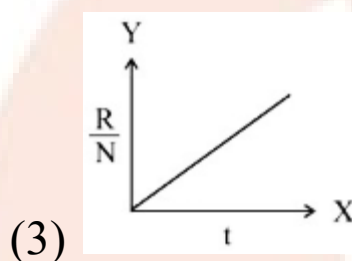
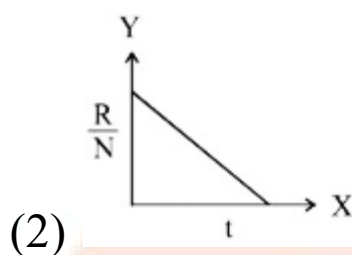
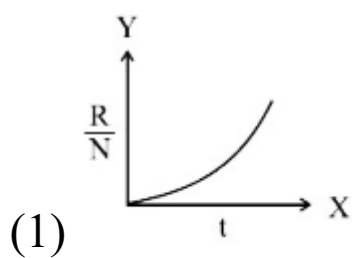
$E = a[1 + \cos(2\pi f_1 t)] \cos 2\pi f_2 t$ (where 'a' is constant) is ($f_1 = 3.6 \times 10^{15}$ Hz and $f_2 = 1.2 \times 10^{15}$ Hz and Planck's constant $h = 6.6 \times 10^{-34}$ Js)

- (1) 2.64 eV
- (2) 7.55 eV
- (3) 12.52 eV
- (4) 17.45 eV

117. Magnetic moment due to the motion of the electron in the n^{th} energy state of hydrogen atom is proportional to _____

- (1) n^{-2}
- (2) n
- (3) n^2
- (4) n^3

118. The rate of disintegration of a radioactive sample is 'R' and the number of atoms present at any time 't' is 'N'. When $\frac{R}{N}$ is taken along Y-axis and 't' is taken along the X-axis, the correct graphs is



119. For an LED to emit light in the visible region of the electromagnetic spectrum, it can have energy band gap in the range of

(Plank's constant $= 6.6 \times 10^{-34}$ Js and speed of light $= 3 \times 10^8$ ms⁻¹ in vacuum)

(1) 0.1 eV to 0.4 eV

(2) 0.9 eV to 1.6 eV

(3) 1.7 eV to 3.1 eV

(4) 0.5 eV to 0.8 eV

120. A transmitting antenna of height 20 m and the receiving antenna of height 'h' are separated by a distance of 40 km for satisfactory communication in the line of sight mode. Then the value of 'h' is

(Given Radius of earth is 6400 km)

(1) 40 m

(2) 45 m

(3) 30 m

(4) 25 m

121. The energies of an electron in first orbit of He^+ and in third orbit of Li^{2+} in J are respectively

(1) -8.72×10^{-18} , -2.18×10^{-18}

(2) -8.72×10^{-18} , -1.96×10^{-17}

(3) -1.96×10^{-17} , -2.18×10^{-18}

(4) -8.72×10^{-17} , -1.96×10^{-17}

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Solutions – Physics

81. The value of T is calculated as,

$$\begin{aligned} T &= \sqrt{(\Delta t_1)(\Delta t_2)} \\ &= \sqrt{(2.00)(4.00)} \\ &= 2.828427 \text{ s} \end{aligned}$$

The round off value of T , so $T = 2.83$ s.

The expression for the relative error of the product is,

$$\pm \frac{\Delta T}{T} = \pm \left(\frac{1}{2} \frac{\Delta t_1}{t_1} + \frac{1}{2} \frac{\Delta t_2}{t_2} \right)$$

Substitute the given values in the above equation.

$$\begin{aligned} \Delta T &= \pm \left(\frac{1}{2} \frac{\Delta t_1}{t_1} + \frac{1}{2} \frac{\Delta t_2}{t_2} \right) T \\ &= \pm \left(\frac{1}{2} \frac{0.02}{2.00} + \frac{1}{2} \frac{0.02}{4.00} \right) \times 2.8284 \\ &= 0.02121 \text{ s} \end{aligned}$$

The round off value of ΔT , so $\Delta T = 0.02$ s

Thus, $T = (2.83 + 0.02)$ s

82. The expression of the angle between the initial and the final velocities of the particle is,

$$\cos \theta = \frac{\Delta v^2 - v_i^2 - v_f^2}{2v_i v_f}$$

Substitute the given values in the above equation.

$$\cos \theta = \frac{(5)^2 - (\sqrt{5})^2 - (2\sqrt{5})^2}{2(\sqrt{5})(\sqrt{5})}$$

$$\cos \theta = 0$$

$$\theta = 90^\circ$$

83. The expression for the maximum height of the projectile is given by,

$$H = \frac{u^2 \sin^2 \theta}{2g} \dots\dots\dots(1)$$

The expression for the maximum range of the projectile is given by,

$$R = \frac{u^2 \sin 2\theta}{g}$$

The ratio of the maximum height to the range of the projectile is calculated as,

$$\begin{aligned} \frac{H}{R} &= \frac{\frac{u^2 \sin^2 \theta}{2g}}{\frac{u^2 \sin 2\theta}{g}} \\ &= \frac{\sin^3 \theta}{2 \sin 2\theta} \end{aligned}$$

Substitute the given values in the above equation.

$$\frac{H}{R} = \frac{\sin^3 \theta}{2 \sin 2\theta}$$

$$\frac{3}{4} = \frac{\sin^3 \theta}{2 \sin 2\theta}$$

$$3 \sin 2\theta = 2 \sin^2 \theta$$

$$\tan \theta = 3$$

From the geometry of the triangle,

$$\sin \theta = \frac{3}{\sqrt{10}} \dots\dots\dots(2)$$

From the equation (1),

$$\frac{u^2 \sin^2 \theta}{2g} = 3$$

$$u^2 \sin^2 \theta = 3 \times 2g$$

Substitute the values in the above equation.

$$u^2 \left(\frac{3}{\sqrt{10}} \right)^2 = 3 \times 2(10)$$

$$u^2 = \frac{100 \times 2}{3}$$

$$u = 10\sqrt{\frac{2}{3}} \text{ ms}^{-1}$$

84. The expression of the displacement from the equation of the motion is,

$$s = ut + \frac{1}{2}gt^2$$

Substitute the values assuming complete flight.

$$0 = u(2) - \frac{1}{2}(10)(2)^2$$

$$u = 10 \text{ ms}^{-1}$$

The expression of the maximum height is,

$$h = ut - \frac{1}{2}gt^2$$

Substitute the values in the above equation.

$$\begin{aligned} h &= (10)(1) - \frac{1}{2}(10)(1)^2 \\ &= 5 \text{ m} \end{aligned}$$

85. From the graph, the final velocity is calculated as,

$$\begin{aligned} v_f &= \frac{3-3}{8-4} \\ &= 0 \end{aligned}$$

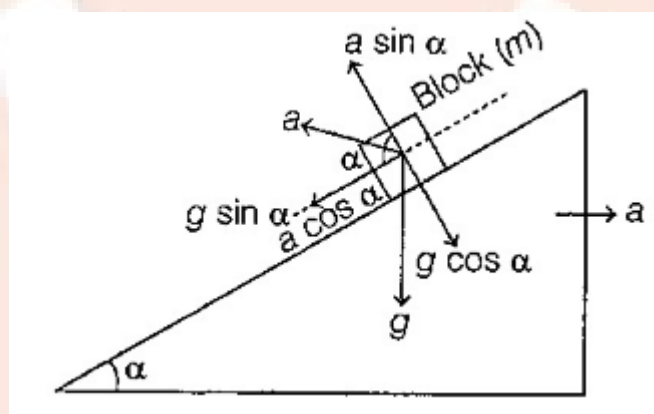
From the graph, the initial velocity is calculated as,

$$\begin{aligned} v_i &= \frac{3-0}{4-0} \\ &= \frac{3}{4} \end{aligned}$$

The impulse is calculated as,

$$\begin{aligned}
 \text{Impulse} &= m(v_f - v_i) \\
 &= 2 \times \left(0 - \frac{3}{4} \right) \\
 &= -1.5 \text{ kg} \cdot \text{ms}^{-1}
 \end{aligned}$$

86. Consider the free body diagram of the block plane system as shown below.



The expression of the downward acceleration is,

$$a_{\text{down}} = g \sin \alpha + a \cos \alpha$$

The expression of the upward acceleration is,

$$a_{\text{up}} = \mu(g \cos \alpha - a \sin \alpha)$$

The condition for the block to remain stationary is,

$$a_{\text{down}} = a_{\text{up}}$$

$$\mu = \frac{g \sin \alpha + a \cos \alpha}{g \cos \alpha - a \sin \alpha}$$

Substitute the given values in above expression,

$$\mu = \frac{10 + 2 \frac{1}{\tan \alpha}}{10 \frac{1}{\tan \alpha} - 2}$$

$$\mu = \frac{10 + 2 \times 5}{10 \times 5 - 2}$$

$$\mu = \frac{20}{48}$$

$$\mu = \frac{5}{12}$$

87. Condition for the minimum potential energy is,

$$U_{\min} = \frac{dU(x)}{dx}$$

$$0 = \frac{d}{dx} \left[\frac{x^2}{2} - x \right]$$

$$0 = x - 1$$

$$x = 1$$

The minimum potential energy is calculated as,

$$\begin{aligned} U_{\min} &= \frac{(1)^2}{2} - 1 \\ &= -\frac{1}{2} \end{aligned}$$

The expression of the kinetic energy is,

$$\begin{aligned}
 KE &= \frac{1}{2}mv^2 \\
 &= \frac{1}{2}(1)v^2 \\
 &= \frac{v^2}{2}
 \end{aligned}$$

The total mechanical energy of the system is given by,

$$E = U + KE$$

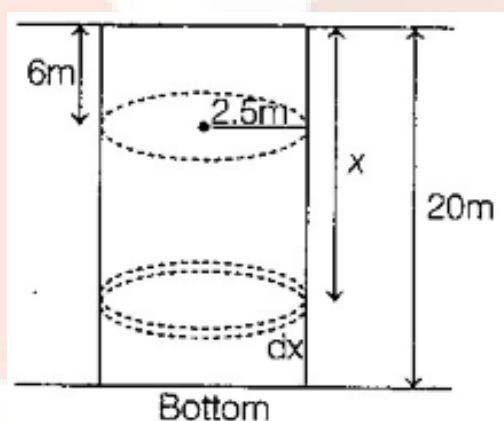
Substitute the given values in the above equation.

$$E = U + KE$$

$$2 = -\frac{1}{2} + \frac{v^2}{2}$$

$$v = \sqrt{5} \text{ m/s}$$

88. Consider the small portion of element dx as shown below.



The mass of the element is,

$$\begin{aligned}
 dm &= \rho dV \\
 &= \rho \pi r^2 dx
 \end{aligned}$$

The potential energy of the mass dm is calculated as,

$$dU = (dm)gx$$

$$dU = g\rho\pi r^2 x dx$$

Integrate the above equation on both sides.

$$U = g\rho\pi r^2 \int_6^{20} x dx$$

Substitute the given values in the above equation.

$$\begin{aligned} U &= 10 \times 10^3 \times 3.14 \times (2.5)^2 \left[\frac{20^2}{2} - \frac{6^2}{2} \right] \\ &= 10^4 \times 3.14 \times (2.5)^2 \times 182 \\ &= 35.71 \times 10^6 \text{ J} \end{aligned}$$

The time taken by the pump is calculated as,

$$\begin{aligned} \text{time} &= \frac{\text{work done}}{\text{power}} \\ &= \frac{35.71 \times 10^6}{10 \times 746 \times 60} \\ &= 80 \text{ min} \end{aligned}$$

89. The magnitude of the radius is calculated as,

$$\begin{aligned} R &= \sqrt{(2)^2 + (1)^2 + (2)^2} \\ &= \sqrt{9} \\ &= 3 \text{ m} \end{aligned}$$

The magnitude of the force is calculated as,

$$\begin{aligned} F &= \sqrt{(3)^2 + (2)^2 + (-4)^2} \\ &= \sqrt{29} \text{ N} \end{aligned}$$

The torque on the flywheel is calculated as,

$$\begin{aligned} \tau &= I\alpha \\ FR &= \left(\frac{MR^2}{2} \right) \alpha \end{aligned}$$

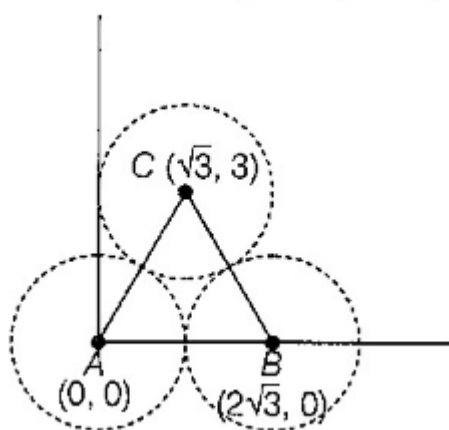
The angular acceleration is,

$$\begin{aligned} \alpha &= \frac{2F}{MR} \\ &= \frac{2\sqrt{29}}{1 \times 3} \\ &= \frac{2}{3}\sqrt{29} \end{aligned}$$

The angular velocity is calculated as,

$$\begin{aligned} \omega &= \omega_0 + \alpha t \\ &= 0 + \frac{2}{3}\sqrt{29} \times (4.5) \\ &= \sqrt{261} \text{ rad s}^{-1} \end{aligned}$$

90. Consider the diagram as shown below,



The expression for the centre of the mass of the spheres is,

$$x_{CM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

The centre of the mass of the spheres along the x axis is calculated as,

$$\begin{aligned} x_{CM} &= \frac{m}{3m} (0 + 2\sqrt{3} + \sqrt{3}) \\ &= \frac{3\sqrt{3}}{3} \\ &= \sqrt{3} \end{aligned}$$

The centre of the mass of the spheres along the y axis is calculated as,

$$\begin{aligned} y_{CM} &= \frac{m}{3m} (0 + 0 + 3) \\ &= 1 \end{aligned}$$

The coordinates of the centre of the mass of the sphere is

$$C_{CM} = (\sqrt{3}, 1).$$

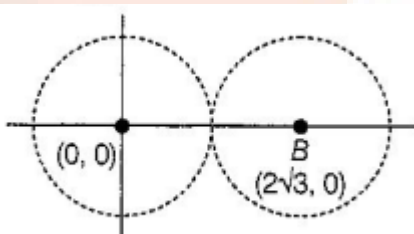
According to the given condition, the centre of sphere when one sphere is removed is,

$$x'_{CM} = \frac{0 + 2\sqrt{3}}{2}$$

$$= \sqrt{3}$$

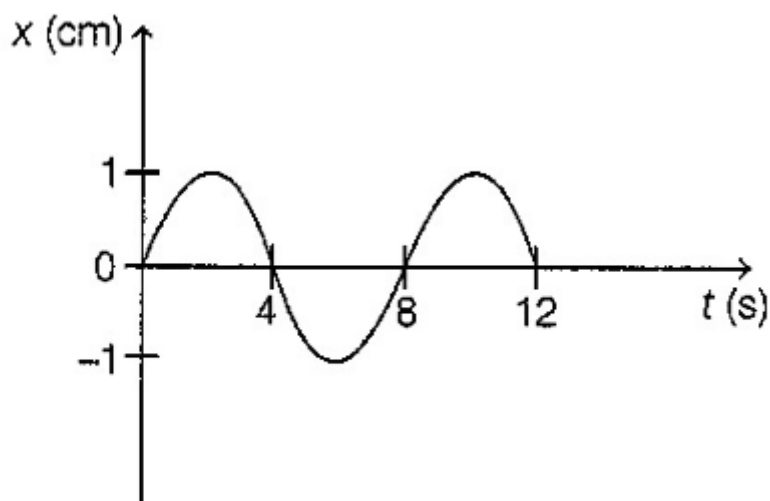
$$y'_{CM} = 0$$

The below diagram is a representation of spheres after the exclusion of one sphere.



The new coordinates of the centre of mass of the spheres is $C'_{CM} = (\sqrt{3}, 0)$, which implies that the centre of the mass is shifted by 1 m in y direction.

91. The diagram of the sine wave function is shown below.



The expression for the angular velocity is given by,

$$\omega = \frac{2\pi}{T}$$

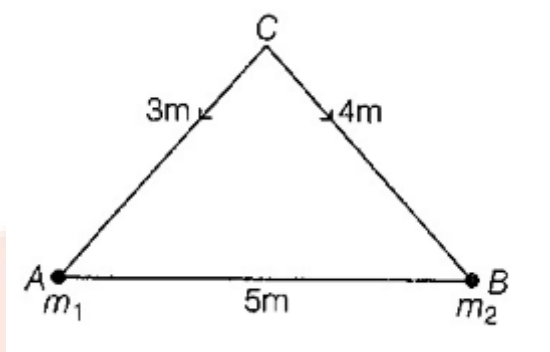
Substitute the given value in the above equation.

$$\begin{aligned}\omega &= \frac{2\pi}{8} \quad (\because T = 8 \text{ s for complete cycle}) \\ &= \frac{\pi}{4} \text{ rad/s}\end{aligned}$$

The acceleration in the simple harmonic motion is calculated as,

$$\begin{aligned}\alpha &= -\omega^2 a \sin \omega t \\ &= -\left(\frac{\pi}{4}\right)^2 \times 1 \times \sin\left(\frac{\pi}{4} \times \frac{4}{3}\right) \\ &= -\frac{\pi^2}{16} \times \frac{\sqrt{3}}{2} \\ &= -\frac{\pi^2 \sqrt{3}}{32} \text{ cm s}^{-2}\end{aligned}$$

92. The below diagram shows the system of the two masses.



The gravitational field intensity due to the mass A is calculated as,

$$\begin{aligned} E_A &= \frac{GM_A}{r_{CA}^2} \\ &= G \frac{90}{(3^2)} \\ &= 10G \hat{r}_{CA} \end{aligned}$$

The gravitational field intensity due to the mass B is calculated as,

$$\begin{aligned} E_B &= \frac{GM_B}{r_{CB}^2} \\ &= G \frac{160}{(4^2)} \\ &= 10G \hat{r}_{CB} \end{aligned}$$

The resultant of E_A and E_B is,

$$\begin{aligned}
 E &= \sqrt{E_A^2 + E_B^2} \\
 &= \sqrt{(10G)^2 + (10G)^2} \\
 &= 10\sqrt{2}G
 \end{aligned}$$

Substitute the value of the G .

$$\begin{aligned}
 E &= 10\sqrt{2}G \\
 &= 10\sqrt{2} \times 6.67 \times 10^{-11} \\
 &= 9.43 \times 10^{-10} \text{ Nkg}^{-1}
 \end{aligned}$$

93. The expression for the extension of the wire is given by,

$$\begin{aligned}
 \Delta L &= \frac{FL}{YA} \\
 \Delta L &= \frac{4F}{Y\pi} \times \frac{L}{D^2} \\
 \Delta L &\propto \frac{L}{D^2}
 \end{aligned}$$

Substitute the values given in the option one by one.

For option (1), the extension is calculated as,

$$\begin{aligned}
 \Delta L &= \frac{4F}{Y\pi} \times \frac{0.5}{(0.5 \times 10^{-3})^2} \\
 &= \frac{4F}{Y\pi} \times 2 \times 10^6
 \end{aligned}$$

Similarly, for option (2), the extension is,

$$\Delta L = \frac{4F}{Y\pi} \times 1.0 \times 10^6$$

For option (3), the extension is,

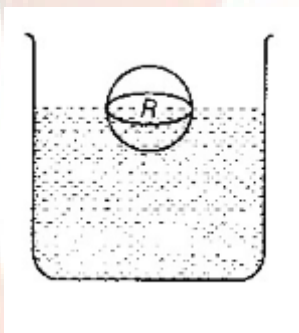
$$\Delta L = \frac{4F}{Y\pi} \times 0.5 \times 10^6$$

For option (4), the extension is,

$$\Delta L = \frac{4F}{Y\pi} \times 0.33 \times 10^6$$

The highest value of extension is associated with the option (1).

94. The liquid drop half immersed in the liquid is shown below.



The force acting on the drop due to surface tension is calculated as,

$$\begin{aligned} F &= Tl \\ &= 10g(\pi D) \end{aligned}$$

The observed weight of the drop is calculated as,

$$\begin{aligned}
 w &= w_d - w_L \\
 &= \rho V g - \frac{\rho}{2} \frac{V}{2} g \\
 &= \frac{3}{4} \rho V g
 \end{aligned}$$

Condition for the equilibrium of ball is,

$$F = w$$

Substitute values,

$$\begin{aligned}
 10g\pi D &= \frac{3}{4} \rho g \frac{\pi D^3}{6} \\
 D &= \sqrt{\frac{80}{\rho}} \text{ m}
 \end{aligned}$$

95. The expression of the Newton's law of cooling is given by,

$$-\frac{dT}{dt} = k'(T - T_0)$$

Integrate the equation on both sides.

$$\int \frac{dT}{T - T_0} = k' \int dt$$

$$\log_e (T - T_0) = -k't$$

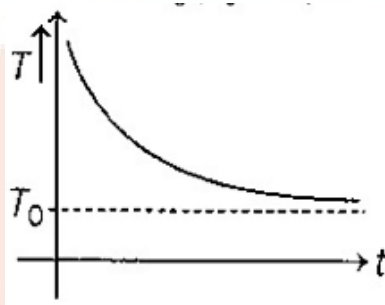
$$\ln(T - T_0) = -k't$$

$$\text{So, } T = e^{-k't} + T_0 \text{ for } t \rightarrow \infty$$

$$\text{And, } T = T_0 \text{ for } T \rightarrow \infty.$$

Hence, the temperature of the body varies exponentially with the time from T to T_0 .

Thus the graph of T verses t is shown below.



96. The expression for the rate of fall of temperature is given by,

$$\frac{\Delta T}{\Delta t} = \frac{\sigma Ae(T^4 - T_0^4)}{ms}$$

$$\Delta t = \frac{ms\Delta T}{\sigma Ae(T^4 - T_0^4)}$$

Substitute the given values in the given equation.

$$\begin{aligned}\Delta t &= \frac{ms\Delta T}{\sigma Ae(T^4 - T_0^4)} \\ &= \frac{(\rho V)C(200\text{K} - 100\text{K})}{\sigma(A)(200^4 - 0^4)} \\ &= \frac{\rho \frac{4}{3} \pi r^3 C \times 100}{\sigma 4\pi r^2 \times (200)^4} \\ &= \frac{1}{48} \frac{\rho r C}{\sigma} \mu\text{s}\end{aligned}$$

97. The expression of the efficiency of the Carnot engine is,

$$\eta = 1 - \frac{T_2}{T_1}$$

For (A), the efficiency is,

$$\begin{aligned}\eta &= 1 - \frac{300}{500} \\ &= 0.4\end{aligned}$$

For (B), the efficiency is,

$$\begin{aligned}\eta &= 1 - \frac{350}{500} \\ &= 0.3\end{aligned}$$

For (C), the efficiency is,

$$\begin{aligned}\eta &= 1 - \frac{400}{800} \\ &= 0.5\end{aligned}$$

For (D), the efficiency is,

$$\begin{aligned}\eta &= 1 - \frac{360}{450} \\ &= 0.8\end{aligned}$$

Thus option (4) is correct.

98. The expression for the kinetic energy is given by,

$$KE = \frac{1}{2}mv^2$$

Substitute the given values in the above equation.

$$\begin{aligned} KE &= \frac{1}{2} \times 200 \times 8^2 \\ &= 6400 \text{ J} \end{aligned}$$

According to the statement of the question,

$$\begin{aligned} H &= \frac{6400 \times 23}{100} \\ &= 1472 \text{ J} \end{aligned}$$

The rise in temperature of the steel is calculated as,

$$\begin{aligned} \Delta T &= \frac{H}{ms} \\ &= \frac{1472}{460 \times 0.2} \\ &= 16 \text{ K} \end{aligned}$$

99. The expression for the mean free path is given by,

$$\bar{\lambda} = \frac{kT}{\sqrt{2}\pi d^2 p} \quad \text{..... (I)}$$

The expression for the frequency is given by,

$$\nu = \frac{v}{\lambda} \quad \text{..... (II)}$$

From equation (I) and (II),

$$\nu = \frac{\sqrt{2}\pi d^2 p \times v}{kT}$$

Substitute the given values in the above equation.

$$\begin{aligned}\nu &= \frac{\sqrt{2}(3.14)(10^{-20})(10^5) \times 1380}{(1.38 \times 10^{-23}) \times 314} \\ &= \sqrt{2} \times 10^9 \text{ Hz} \\ &= 1000\sqrt{2} \text{ MHz}\end{aligned}$$

100. The variation of the speed of sound with the temperature is given by,

$$\nu_t = \nu_0 \sqrt{\frac{T}{T_0}}$$

Substitute the given values in the above equation.

$$\begin{aligned}\nu_t &= \nu_0 \sqrt{\frac{330.75}{300}} \\ &= 1.05\nu_0\end{aligned}$$

The number of the beats is calculated as,

$$\begin{aligned}n &= n_t - n_0 \\ &= 140(1.05) - 140 \\ &= 7\end{aligned}$$

101. The expression for the apparent frequency for the moving source is,

$$n = \frac{n_0 v}{v - v_s}$$

The condition for $n_{\max}$, $v_s = v_{s\max}$ and for $n_{\min}$, $v_s = -v_{s\max}$.

The expression for $n_{\max}$ is given by,

$$n_{\max} = \frac{n_0 v}{v - v_{s\max}} \dots (1)$$

The expression for $n_{\min}$ is given by,

$$n_{\min} = \frac{n_0 v}{v + v_{s\max}} \dots (2)$$

Divide equation (1) by (2).

$$\frac{1.125 n_{\min}}{n_{\min}} = \frac{v + v_{s\max}}{v - v_{s\max}}$$

$$1.125(v - v_{s\max}) = v + v_{s\max}$$

$$v(0.125) = 2.125 v_{s\max}$$

$$\begin{aligned} v_{s\max} &= \frac{340 \times 0.125}{2.125} \\ &= 20 \text{ ms}^{-1} \end{aligned}$$

The expression for the $v_{s\max}$ is given by,

$$v_{s_{\max}} = A \sqrt{\frac{k}{m}}$$

$$k = \frac{v_{s_{\max}}^2 m}{A^2}$$

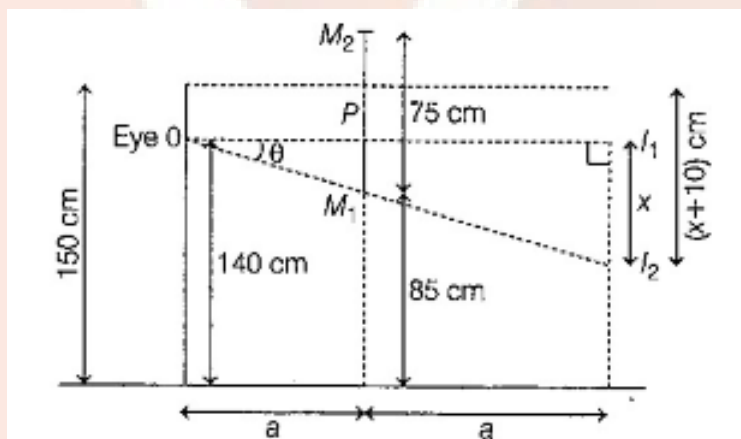
Substitute the given values in the above equation.

$$k = \frac{v_{s_{\max}}^2 m}{A^2}$$

$$= \frac{20 \times 20 \times 0.10}{0.50 \times 0.50}$$

$$= 160 \text{ N m}^{-1}$$

102. The position of a girl in front of a plane mirror is shown below.



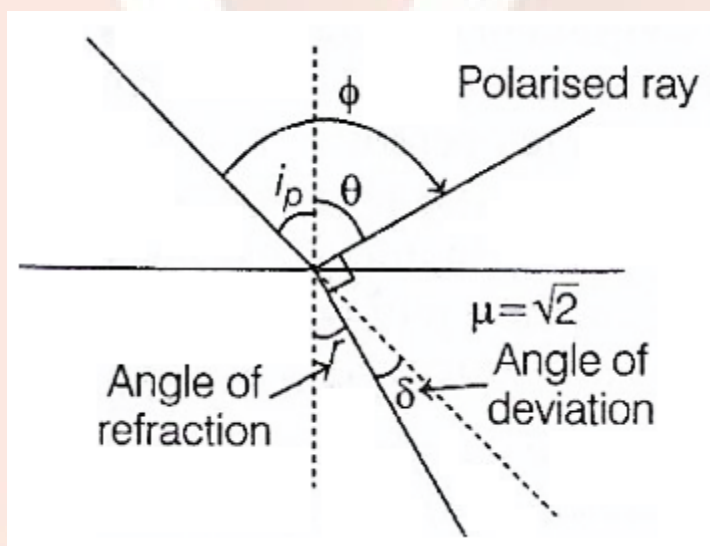
From the property of the triangle,

$$\begin{aligned}\tan \theta &= \frac{PM_1}{OP} \\ &= \frac{I_1 I_2}{OP_1} \\ \frac{x}{2a} &= \frac{140 - 85}{a} \\ x &= 110 \text{ cm}\end{aligned}$$

The maximum height seen by the girl is calculated as,

$$\begin{aligned}H &= 110 + 10 \\ &= 120 \text{ cm}\end{aligned}$$

103. The ray diagram of refraction at a plane surface is shown below.



The value of i_p is calculated as,

$$\begin{aligned}i_p &= \tan^{-1} \mu \\&= \tan^{-1} \sqrt{2} \\&= \sin^{-1} \sqrt{\frac{2}{3}}\end{aligned}$$

From Snell's law,

$$\sin i_p = \mu \sin r$$

$$\sin r = \frac{\sin i_p}{\mu}$$

Substitute the given values in the above equation.

$$\sin r = \frac{\sin i_p}{\mu}$$

$$r = \frac{1}{\sqrt{2}} \sqrt{\frac{2}{3}}$$

$$r = \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

The angle of the deviation is calculated as,

$$\delta = i_p - r$$

$$= \sin^{-1} \sqrt{\frac{2}{3}} - \sin^{-1} \frac{1}{\sqrt{3}}$$

The angle of reflection is,

$$\theta = 90^\circ - r$$

$$r = 90^\circ - \theta$$

$$\sin^{-1} \frac{1}{\sqrt{3}} = 90^\circ - \theta$$

$$\frac{1}{\sqrt{3}} = \sin(90^\circ - \theta)$$

Solve further,

$$\cos \theta = \frac{1}{\sqrt{3}}$$

$$\theta = \cos^{-1} \frac{1}{\sqrt{3}}$$

The angle between incident and completely polarised light is calculated as,

$$\phi = i_p + \theta$$

$$= \sin^{-1} \sqrt{\frac{2}{3}} + \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$= 2 \sin^{-1} \sqrt{\frac{2}{3}}$$

104. The expression for the electric field intensity on the axis is given by,

$$E_{\text{axis}} = \frac{2kP}{r^3}$$

$$4 = \frac{2kP}{r^3}$$

$$\frac{kP}{r^3} = 2$$

The expression for the electric field intensity on the equatorial axis is given by,

$$E_{eq} = \frac{k'P}{r_1^3}$$

Here, $r_1 = 2r$ and $k' = \frac{k}{4}$

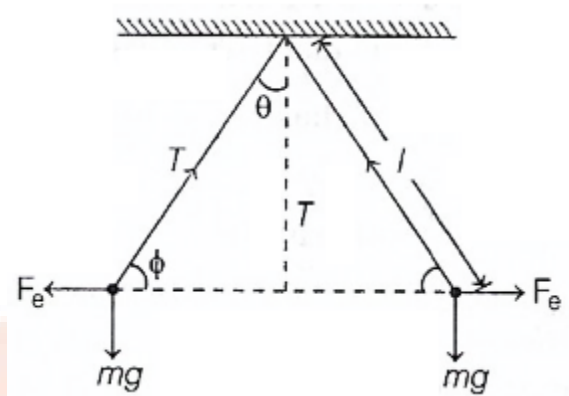
So,

$$E_{eq} = \frac{kP}{4 \times 8r^3}$$

Substitute 2 for $\frac{kP}{r^3}$ in the above equation.

$$\begin{aligned} E_{eq} &= \frac{2}{4 \times 8} \\ &= \frac{1}{16} \text{ NC}^{-1} \end{aligned}$$

105. Consider the below diagram.



$$F_e = \frac{kq^2}{r^2} \dots (1)$$

$$r = 2l \cos \theta$$

$$\tan \theta = \frac{F_e}{mg}$$

For the value of $\theta = 45^\circ$ and $\phi = 45^\circ$.

$$\begin{aligned} r &= \frac{2l}{\sqrt{2}} \\ &= \sqrt{2}l \end{aligned}$$

Substitute the value of r in equation (1).

$$F_e = \frac{kq^2}{2l}$$

From the geometry of the triangle,

$$\tan 45^\circ = \frac{kq^2}{2lmg}$$

$$mg = \frac{kq^2}{2l} \dots (2)$$

For the value of $\theta = 30^\circ$ and $\phi = 60^\circ$.

$$r = 2l \times \frac{1}{2}$$

$$= l$$

Similarly,

$$F'_e = \frac{kq^2}{l^2}$$

So,

$$\tan 30^\circ = \frac{k'q^2}{l^2 - m'g}$$

Assuming that the sphere is suspended in a liquid of density $\frac{2}{3}d$

.

The observed weight of the body is calculated as,

$$m' = V \left(d - \frac{2d}{3} \right)$$

$$= \frac{m}{3}$$

Thus,

$$\tan 30^\circ = \frac{3k'q^2}{l^2 mg}$$

$$mg = \frac{3\sqrt{3}k'q^2}{l^2} \dots (3)$$

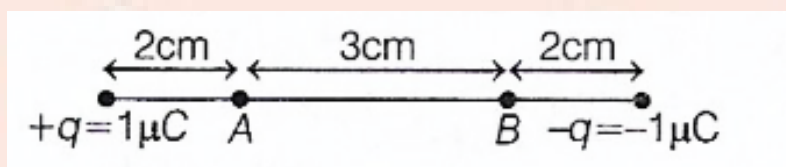
From equation (2) and (3),

$$3\sqrt{3}k' = \frac{k}{2}$$

$$k' = \frac{k}{\epsilon_r}$$

$$\epsilon_r = 6\sqrt{3}$$

106. Consider the below diagram.



The potential at the point A is calculated as,

$$\begin{aligned} V_A &= V_{A^{+q}} + V_{A^{-q}} \\ &= \frac{9 \times 10^9 \times 10^{-6}}{2 \times 10^{-2}} - \frac{9 \times 10^9 \times 10^{-6}}{5 \times 10^{-2}} \\ &= 9 \times 10^5 \left[\frac{1}{2} - \frac{1}{5} \right] \end{aligned}$$

The potential at B is calculated as,

$$\begin{aligned} V_B &= V_{B^{+q}} + V_{B^{-q}} \\ &= 9 \times 10^5 \left[\frac{1}{5} - \frac{1}{2} \right] \end{aligned}$$

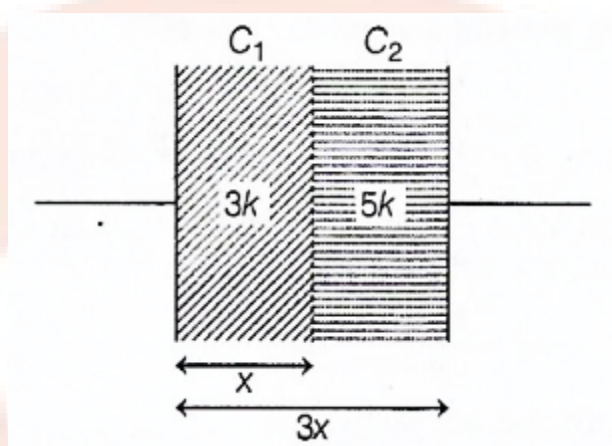
The potential difference is calculated as,

$$V_A - V_B = 5.4 \times 10^5 \text{ V}$$

107. The expression for the capacitance of a parallel plate is,

$$C = \epsilon_r \cdot \frac{\epsilon_0 A}{d}$$

The capacitor is shown below.



Here, $d_0 = 3x$, $d_1 = x$, $\epsilon_1 = 3k$, $d_2 = 2x$ and $\epsilon_2 = 5k$

The capacitance across the plates is calculated as,

$$C_1 = 3k \frac{\epsilon_0 A}{x} \text{ and } C_2 = 5k \frac{\epsilon_0 A}{2x}$$

For the combination of the capacitor in the series.

$$V_1 = \frac{Q}{C_1} \text{ and } V_2 = \frac{Q}{C_2}$$

Substitute the values in the above equation.

$$V_1 = \frac{\theta x}{3k \epsilon_0 A} \text{ and } V_2 = \frac{\theta 2x}{5k \epsilon_0 A}$$

The ratio of the potential difference is calculated as,

$$\begin{aligned}
 \frac{V_1}{V_2} &= \frac{\frac{\theta x}{3k\epsilon_0 A}}{\frac{\theta 2x}{5k\epsilon_0 A}} \\
 &= \frac{5}{3 \times 2} \\
 &= \frac{5}{6}
 \end{aligned}$$

108. The conductivity of the metal decreases with the increase in the temperature whereas the conductivity of semiconductor increases with the increase in the temperature.

So, the conductivity of aluminium decreases with the increase in the temperature whereas the conductivity of the silicon increases with the increase in the temperature. Hence, assertion (A) is incorrect but the reason (R) is correct.

109. The expression for the heat transfer through the conduction wall is given by,

$$\frac{dQ}{dt} = kA \frac{(T_1 - T_0)}{x}$$

Substitute the given values in the above equation.

$$\begin{aligned}
 P &= \frac{kA \times 4.184 \times (T_1 - T_0)}{x} \\
 &= \frac{k6a^2 \times 4.184 \times 10^3}{x} \\
 &= \frac{4 \times 10^{-4} \times 6 \times (60^2) \times 10^3 \times 4.184}{0.1} \\
 &= 361.49 \text{ W}
 \end{aligned}$$

The resistance is calculated as,

$$\begin{aligned}
 P &= \frac{V^2}{R} \\
 R &= \frac{V^2}{P} \\
 &= \frac{400 \times 400}{361.49 \times 10^3} \\
 &= 0.4426 \Omega
 \end{aligned}$$

110. The expression of the net resistance of circuit is,

$$R_0 = R + \frac{G \times S}{G + R} \dots (1)$$

According to the given condition,

$$R_0 = G \dots (2)$$

From the equation (1) and (2),

$$G = R + \frac{G \times S}{G + S}$$

$$R = G - \frac{GS}{G + S}$$

$$R = \frac{G^2}{G + S}$$

111. The expression for the time period of revolution of proton in the magnetic field is given by,

$$T_p = \frac{2\pi m_p}{eB} \dots (1)$$

The expression for the time period of revolution of α particle in the magnetic field is given by,

$$T_\alpha = \frac{2\pi m_p}{qB} \dots (2)$$

The ratio of the time period of the proton and α particle is calculated as,

$$\frac{T_p}{T_\alpha} = \frac{1}{2}$$

$$T_\alpha = 2T_p$$

112. The expression for the torques in the magnetic field is given by,

$$\tau = MB \sin \theta$$

For $\theta = \theta_1$.

$$\tau_1 = MB \sin \theta_1 \dots (1)$$

Similarly,

$$\begin{aligned}\tau_2 &= MB \sin \theta_2 \\ &= MB \sin 2\theta_1\end{aligned}$$

According to the given condition,

$$\begin{aligned}\tau_2 &= \tau_1 + \tau_1 \times \frac{41.4}{100} \\ &= 1.414\tau_1 \\ &= \sqrt{2}\tau_1\end{aligned}$$

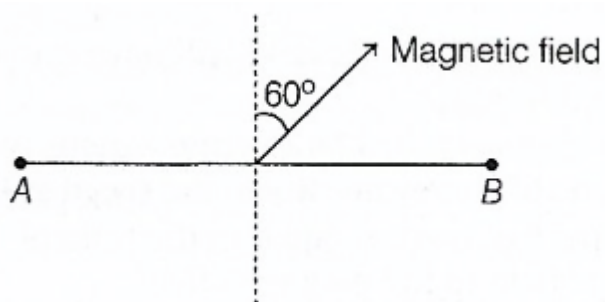
So,

$$\sqrt{2}\tau_1 = MB \sin 2\theta_1 \dots (2)$$

From equation (1) and (2),

$$\begin{aligned}\frac{1}{\sqrt{2}} &= \frac{\sin \theta_1}{\sin 2\theta_1} \\ \sin 2\theta_1 &= \sqrt{2} \sin \theta_1 \\ 2 \sin \theta_1 \cos \theta_1 &= \sqrt{2} \sin \theta_1 \\ \cos \theta_1 &= \frac{\sqrt{2}}{2} \\ \theta_1 &= 45^\circ\end{aligned}$$

113. Consider the below diagram.



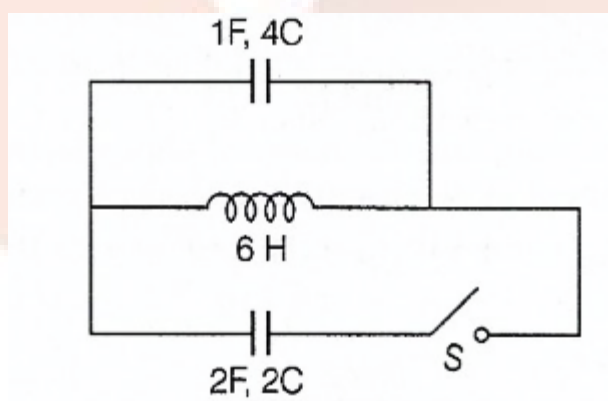
The expression for the emf induced in the metal rod is given by,

$$\begin{aligned} e_{AB} &= V_A - V_B \\ &= l \cdot (\mathbf{v} \times \mathbf{B}) \\ &= lvB \sin(90^\circ - 60^\circ) \end{aligned}$$

Substitute the given values in the above equation.

$$\begin{aligned} V_A - V_B &= 50 \times 10^{-2} \times 8 \times 2 \times \sin 30^\circ \\ &= 8 \times \frac{1}{2} \\ &= 4 \text{ V} \end{aligned}$$

114. Consider the below arrangement of the electrical circuit.



The energy in 1F capacitor is calculated as,

$$\begin{aligned}E_1 &= \frac{1}{2}CV^2 \\&= \frac{1}{2}C\left(\frac{Q}{C}\right)^2 \\&= \frac{1}{2}\frac{Q^2}{C} \\&= \frac{1}{2}\times\frac{4^2}{1} \\&= 8 \text{ J}\end{aligned}$$

Similarly,

$$\begin{aligned}E_2 &= \frac{1}{2}\frac{Q^2}{C} \\&= \frac{2^2}{2\times 2} \\&= 1 \text{ J}\end{aligned}$$

The total energy is calculated as,

$$\begin{aligned}E_0 &= E_1 + E_2 \\&= 8 + 1 \\&= 9 \text{ J}\end{aligned}$$

The common potential of the capacitors is calculated as,

$$\begin{aligned}V_{\text{common}} &= \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} \\&= \frac{1 \times 4 + 2 \times 1}{1 + 2} \\&= \frac{6}{3} \\&= 2 \text{ V}\end{aligned}$$

For the new arrangement of the circuit the energy is calculated as,

$$\begin{aligned}E_1 &= \frac{1}{2} C_1 V_{\text{common}}^2 \\&= \frac{1}{2} \times 1 \times 4 \\&= 2 \text{ J}\end{aligned}$$

Similarly,

$$\begin{aligned}E_2 &= \frac{1}{2} C_2 V_{\text{common}}^2 \\&= \frac{1}{2} \times 2 \times 4 \\&= 4 \text{ J}\end{aligned}$$

From the conservation of the energy, the energy stored in the inductor is calculated as,

$$\begin{aligned}E_L &= E_0 - (E_1 + E_2) \\&= 9 - (2 + 4) \\&= 3 \text{ J}\end{aligned}$$

115. The energy of the electromagnetic wave is equally distributed between electric and magnetic field.

Thus, option (2) is correct.

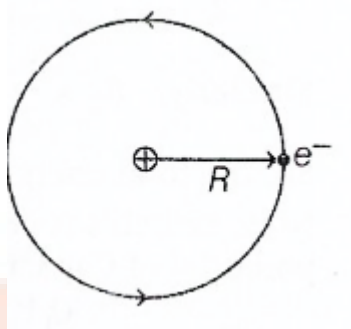
116. The maximum energy of photoelectron is calculated as,

$$\begin{aligned}E_{\max} &= \frac{h\nu_{\max}}{e} \\&= \frac{6.6 \times 10^{-34} \times (3.6 \times 10^{15} + 1.2 \times 10^{15})}{1.6 \times 10^{-19}} \\&= 19.8 \text{ eV}\end{aligned}$$

The maximum kinetic energy is calculated as,

$$\begin{aligned}KE_{\max} &= E_{\max} - W_0 \\&= 19.8 - 2.35 \\&= 17.45 \text{ eV}\end{aligned}$$

117. The magnetic movement of the electron is shown below.



The expression for the magnetic moment is,

$$M = iA$$

$$= \frac{e\omega}{2\pi} \pi R^2$$

Substitute the value of current in the above equation.

$$M = \frac{ev}{2R} R^2$$

$$= \frac{evR}{2}$$

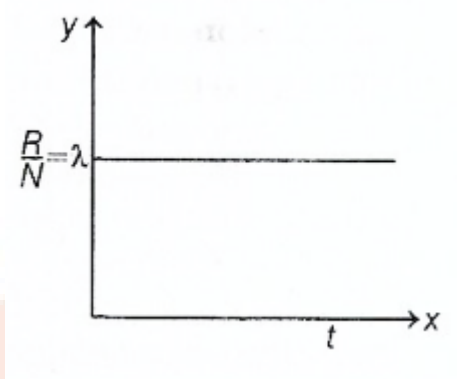
$$= \frac{eL}{2m}$$

Simplify further using the relations $L = mvr = \frac{nh}{2\pi}$.

$$M = \frac{enh}{2\pi m}$$

$$M \propto n$$

118. Consider the below diagram.



The statement of the law of the radioactive decay is,

$$\frac{dN}{dt} = -\lambda N$$

$$R = -\lambda N$$

$$R = -\lambda$$

Thus the correct graph is shown in option (4).

119. The maximum energy of the photon on the visible spectrum is calculated as,

$$E_{\max} = h\nu$$

$$= \frac{6.6 \times 10^{-34} \times 7 \times 10^{14}}{1.6 \times 10^{-19}}$$

$$= 2.88 \text{ eV}$$

The minimum energy of the photon on the visible spectrum is calculated as,

$$\begin{aligned}
 E_{\max} &= h\nu \\
 &= \frac{6.6 \times 10^{-34} \times 4 \times 10^{14}}{1.6 \times 10^{-19}} \\
 &= 1.65 \text{ eV}
 \end{aligned}$$

Which implies the correct range of the energy is 1.7 eV to 3 eV

.

120. The expression of the maximum distance of line of sight mode is,

$$d_m = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Substitute the given value in the above equation.

$$40 = \sqrt{2 \times 6.4 \times 10^3 \times 20 \times 10^{-3}} + \sqrt{2 \times 6.4 \times 10^3 \times h_R \times 10^{-3}}$$

$$40 = 16 + \sqrt{2 \times 6.4 \times 10^3 \times 10^{-3} \times h_R}$$

$$2 \times 6.4 \times h_T = 24 \times 24$$

$$h_T = 45 \text{ m}$$

